

INFORMATION SEARCH AND ANALYSIS SKILL

**“The role of CIA triad in Cloud Computing, DBaaS,
and Big Data.”**



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COMPUTING AND INFORMATION
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PREFACE

Praise be to Allah SWT, the Almighty and Most Merciful, for His blessings and guidance, which have enabled the writer to complete this paper entitled “**The role of CIA triad in Cloud Computing, DBaaS, and Big Data.**” This paper is written as part of an academic requirement and aims to provide basic understanding of data security issues in databases, cloud computing, and big data environments.

During the preparation of this paper, the writer gained new insights not only into common security threats but also into the importance of the CIA Triad confidentiality, integrity, and availability as a fundamental concept in information security. It is expected that this paper can help readers, especially students in the field of information technology, to better understand the challenges and basic security mechanisms used to protect data in modern systems.

The writer would like to express sincere gratitude to the lecturers who have provided guidance and knowledge during the learning process, as well as to friends and colleagues who offered support, discussions, and motivation throughout the completion of this paper.

The writer realizes that this paper is still far from perfect. Therefore, constructive suggestions and feedback are highly appreciated for future improvement. Hopefully, this paper can be useful for both the writer and the readers as part of a continuous learning process in the field of information technology.

Depok, February 2026

Author

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CHAPTER I

INTRODUCTION

1.1 Background

In today's digital era, data has become one of the most valuable assets for organizations, governments, and individuals. Databases store critical operational and personal information, cloud computing enables flexible and scalable service delivery, and big data technologies allow massive volumes of data to be analyzed for strategic decision-making. However, the increasing dependence on these technologies has also expanded the attack surface for security threats, making data protection a critical concern.

Security threats such as SQL injection, unauthorized access, data leakage, account hijacking, malicious insiders, and denial-of-service attacks pose serious risks to information systems. These threats are further amplified by the use of multi-tenant architectures in cloud computing, the involvement of third-party service providers, and the complexity of managing large-scale and heterogeneous big data environments. Human factors, including misconfiguration and lack of security awareness, also contribute significantly to security incidents.

To address these challenges, information security is commonly analyzed using the CIA Triad framework. Confidentiality focuses on preventing unauthorized access to sensitive data, integrity ensures that data remains accurate and unaltered, and availability guarantees that data and services remain accessible when needed. This framework provides a structured approach to evaluating security risks and implementing appropriate countermeasures across different technological domains.

This paper presents a literature-based analysis of security threats and challenges in databases, cloud computing, and big data systems, highlighting how the CIA Triad can be applied as a unifying security model. By examining existing research and mitigation strategies, this study aims to contribute to a clearer understanding of how organizations can strengthen their security posture in an increasingly data-driven world.

1.2 Objective

The purpose of this paper is to explain basic security concepts in databases, cloud computing, and big data using the CIA Triad, which includes confidentiality, integrity, and availability. This paper discusses how data is protected from unauthorized access, data changes, and system downtime through common security mechanisms. In addition, this study briefly explains how security threats are detected and handled using methods such as access control, encryption, authentication, and system monitoring. Through this discussion, the paper aims to help readers understand the importance of data security in modern information systems.

1.3 Problem Domain

This paper focuses on data security problems in databases, cloud computing, and big data systems. As the use of digital systems increases, risks such as unauthorized access, data leakage, data modification, and system downtime also become more common. These issues are caused by factors like weak security controls, human error, system misconfiguration, and cyberattacks. Therefore, this paper analyzes these problems using the CIA Triad confidentiality, integrity, and availability as a basic framework to understand data security challenges in modern information systems.

1.4 Writing Methodology

The writing method for this paper uses short journals and literature studies, the author collects information from various reliable sources such as cybersecurity articles, journals, and related websites

1.5 Writing Framework

Systematics that the author uses in this ISAS investigation, is arranged as follows

CHAPTER I: INTRODUCTION

Describes the background, objectives, scope, methodology, and writing framework of the paper.

CHAPTER II: BASIC THEORY

CHAPTER III: PROBLEM ANALYSIS

CHAPTER IV: CONCLUSION AND SUGGESTION

CHAPTER II BASIC THEORY

2.1 CIA Triad

The CIA Triad (Confidentiality, Integrity, and Availability) serves as the fundamental model for information security policies. In the context of modern infrastructure, this framework acts as the primary metric to evaluate the resilience of Cloud, DBaaS, and Big Data systems against the threats identified in the previous chapter.

- **Confidentiality:** Focuses on restricting data access to authorized entities only. It involves implementing encryption and strict access controls to prevent data leakage and unauthorized exposure.
- **Integrity:** Guarantees the trustworthiness and accuracy of data throughout its lifecycle. It ensures that information is not tampered with, altered, or corrupted by unauthorized users or system errors, which is critical in distributed systems.
- **Availability:** Ensures that data and services remain accessible to authorized users whenever needed. This encompasses system resilience against hardware failures, software crashes, and malicious disruptions like Denial of Service (DoS) attacks.

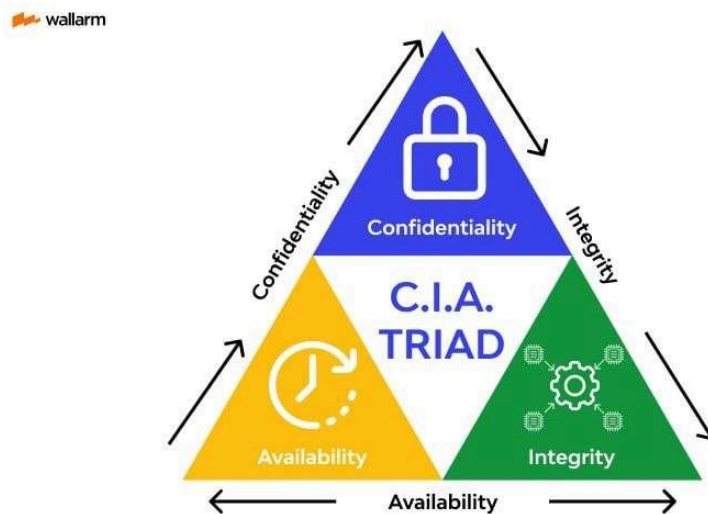


Figure 2.1 CIA Triad (REFF: Wallarm)

2.2 Cloud Computing Security Boundaries

Cloud computing has revolutionized IT by offering unparalleled flexibility, scalability, and cost efficiency. However, from a security perspective, cloud computing introduces a unique challenge that is the Shared Responsibility Model.

Unlike traditional on-premise infrastructure where the organization controls the entire stack, cloud security is a shared obligation between the provider and the user.

- **Provider Responsibility (Security of the Cloud):** The provider secures the underlying infrastructure, physical hardware, and networking logic.
- **User Responsibility (Security in the Cloud):** The organization is responsible for securing the data, configuring access management (IAM), and encrypting traffic.

This model explains why misconfiguration remains a top threat while relying on third-party services creates a complex security landscape where a lack of user-side security controls can expose the entire environment, despite the provider's robust infrastructure.



Figure 2.2 Cloud Computing Illustration (REF: Nevacloud)

2.3 Database as a Service (DBaaS) Architecture Risk

Database as a Service (DBaaS) represents the convergence of cloud scalability with data management, allowing organizations to offload physical maintenance to third-party providers. While efficient, this abstraction layer introduces specific security implications regarding the CIA Triad :

- **Loss of Low-Level Control:** Because the database is hosted in a shared environment, users lose direct control over the operating system and physical storage. This forces reliance on the provider's mechanism to maintain Integrity and Availability.
- **API Vulnerability:** DBaaS relies heavily on Application Programming Interfaces (APIs) for access and management. These interfaces become critical attack vectors; insecure APIs can directly compromise the confidentiality of the hosted database if not secured properly.
- **Data Residency & Backup:** The convenience of automated backups introduces a risk of "unmanaged sensitive data" if these backups are left unencrypted, creating a bypass for attackers to access data without touching the main database.

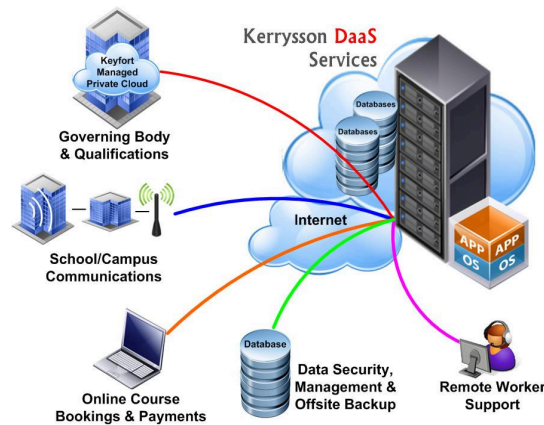


Figure 2.3 DBaaS Illustration (REF: Kerrysson)

2.4 Big Data Challenges

Big Data is defined by datasets so massive and complex that traditional management tools cannot handle them. The security of Big Data is directly challenged by its three defining characteristics :

- **Volume (Challenge to Confidentiality):** Refers to the sheer amount of data. Encrypting and decrypting petabytes of data at rest and in transit requires immense computational power, often leading organizations to skip encryption for performance, thus risking confidentiality.
- **Velocity (Challenge to Integrity):** Refers to the speed of data generation. When data flows in real-time from thousands of sources, validating the integrity of every data packet creates latency. This makes it difficult to detect "data poisoning" or corruption instantly.
- **Variety (Challenge to Availability):** Refers to unstructured formats (logs, video, text). Managing permissions and indexing for such diverse data types is complex, which can lead to data swamps where information exists but is effectively unavailable due to poor organization.



Figure 2.4 Big Data Illustration (REF: iStock)

CHAPTER III

PROBLEM ANALYSIS

3.1 Cloud Computing

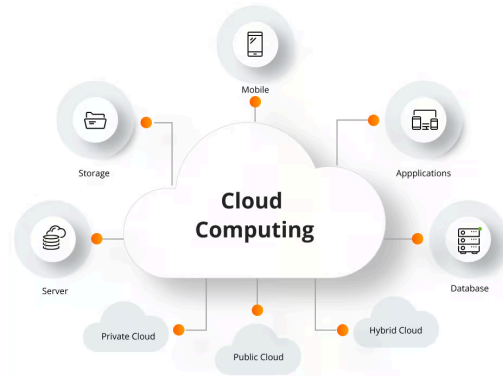


Figure 3.1 Cloud Computing Future Processing

Confidentiality Issues in Cloud Computing

One of the most common confidentiality problems in cloud computing is data leakage caused by misconfiguration. Organizations often misconfigure cloud storage services, access permissions, or security policies, unintentionally exposing sensitive data to the public internet. This issue is amplified by the shared responsibility model, where users are responsible for configuring security correctly.

In addition, insecure APIs and interfaces are a major entry point for unauthorized access. Since cloud services rely heavily on APIs for management and integration, weak authentication or poor API security can lead to serious data breaches. Insider threats either from cloud provider staff or internal employees also pose significant confidentiality risks.

Integrity Issues in Cloud Computing

Maintaining data integrity in cloud environments is challenging due to their distributed and highly integrated architecture. Data can be altered through insecure APIs, compromised credentials, or malicious insiders without being immediately detected.

Furthermore, cloud systems often integrate multiple third-party services. If one component is compromised, it may affect the consistency and correctness of data across the system. Weak logging and insufficient audit mechanisms further increase the risk of undetected data manipulation.

Availability Issues in Cloud Computing

Cloud services are highly dependent on network connectivity and provider infrastructure. As a result, availability is frequently threatened by Denial of Service (DoS/DDoS) attacks, large-scale outages, or provider-side failures.

Even though cloud providers offer redundancy and high availability mechanisms, misconfigured backup systems or inadequate disaster recovery planning on the user side can still lead to service disruptions and data inaccessibility.

3.2 Database as a Service (DBaaS)



Database as a Service

Figure 3.2 Database as as Service on Vmware Cloud

Confidentiality Issues in DBaaS

DBaaS platforms are particularly vulnerable to SQL Injection attacks and weak access control mechanisms. Poorly designed queries or improper input validation can allow attackers to extract sensitive data directly from databases.

Additionally, improper credential management (e.g., hard-coded passwords, exposed API keys) increases the risk of unauthorized database access. Since DBaaS often stores highly sensitive information, confidentiality failures can have severe consequences.

Integrity Issues in DBaaS

Integrity issues in DBaaS commonly arise from misuse of privileged access and insufficient auditing mechanisms. Authorized users may intentionally or unintentionally modify data beyond their intended scope.

In multi-tenant DBaaS environments, transaction conflicts and weak isolation between tenants can also lead to data inconsistency. Without proper audit trails, detecting unauthorized changes becomes difficult.

Availability Issues in DBaaS

DBaaS availability is affected by service outages, maintenance downtime, and dependency on cloud infrastructure. If replication and failover mechanisms are not properly configured, database services may become unavailable during peak usage or system failures.

Because applications rely heavily on continuous database access, even short downtime can significantly impact business operations

3.3 Big Data



Figure 3.3 Data Engineer

Confidentiality Issues in Big Data

Big Data systems process massive volumes of distributed and heterogeneous data, making access control and encryption difficult to manage at scale. Applying fine-grained access policies across multiple nodes and datasets is complex and error-prone.

Moreover, data often originates from multiple sources, increasing the risk of sensitive information exposure if proper anonymization or data masking techniques are not applied.

Integrity Issues in Big Data

Big Data analytics relies on complex data pipelines, including ingestion, transformation, and analysis. Integrity can be compromised if data is corrupted at any stage, especially when validation mechanisms are insufficient.

Because Big Data systems replicate data across nodes, corrupted or manipulated data can propagate quickly, affecting analytical outcomes and decision-making processes.

Availability Issues in Big Data

Big Data infrastructures are designed to tolerate failures, yet node failures, network latency, and cluster mismanagement can still reduce availability. Ensuring continuous access to large-scale data requires effective fault tolerance, replication strategies, and monitoring.

If disaster recovery mechanisms are poorly implemented, restoring access to data may take significant time, impacting real-time analytics and services.

CHAPTER IV

CONCLUSION AND SUGGESTION

4.1 Conclusion

Across Cloud Computing, Database as a Service (DBaaS), and Big Data environments, the CIA Triad—Confidentiality, Integrity, and Availability—remains the gold standard of information security. Despite differences in architecture and use cases, all three technologies share a common dependency on secure data handling, making the CIA Triad universally applicable.

This study demonstrates that even minor failures in one component of the CIA Triad can result in severe and cascading consequences. A small misconfiguration in cloud access control may lead to large-scale data leakage, weak integrity controls in DBaaS can enable unauthorized data manipulation, and limited availability in Big Data systems can disrupt critical analytics and decision-making processes. These risks are amplified by the distributed, scalable, and highly interconnected nature of modern systems.

Furthermore, the findings highlight that security responsibility is not solely technical but also operational and human-centered. Mismanagement, insufficient monitoring, weak auditing, or lack of security awareness can undermine even advanced security infrastructures. As shown across all three objects, confidentiality, integrity, and availability are tightly interconnected—a failure in one dimension often compromises the others.

Therefore, the CIA Triad should not be viewed as a theoretical concept alone, but as a practical and mandatory security baseline. Organizations must implement it consistently and holistically. Any deviation, even seemingly minor, from this standard can lead to fatal security incidents, including data breaches, loss of trust, service disruption, and long-term operational damage.

4.2 Suggestion

Based on the analysis of Cloud Computing, Database as a Service (DBaaS), and Big Data environments, it is strongly recommended that organizations adopt the CIA Triad—Confidentiality, Integrity, and Availability—as the gold standard for information security implementation. The CIA Triad should serve as a mandatory baseline rather than an optional guideline in designing, deploying, and managing modern information systems.

First, organizations should implement confidentiality controls by default, including strong access control policies, encryption of data at rest and in transit, and secure identity and credential management. Regular configuration audits and security assessments are essential to minimize human error and misconfiguration, which have been identified as primary causes of data leakage across cloud-based systems.

Second, to ensure data integrity, organizations must enforce strict privilege management, comprehensive logging, and continuous monitoring mechanisms. Audit trails should be immutable and regularly reviewed to detect unauthorized or accidental data modification. In distributed environments such as DBaaS and Big Data platforms, integrity verification mechanisms should be integrated across all data pipelines and storage layers.

Third, maintaining high availability requires proper redundancy, failover mechanisms, and disaster recovery planning. Organizations should not rely solely on cloud provider availability guarantees but must actively configure and test backup and recovery procedures. Periodic stress testing and incident response simulations are recommended to ensure system resilience under failure or attack scenarios.

Finally, security awareness and governance should be strengthened to ensure that the CIA Triad is consistently applied across technical, operational, and human factors. Continuous training, clear responsibility allocation, and policy enforcement are critical to preventing minor security oversights that could escalate into severe and fatal security incident.

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